Practice Set 8

MA628 - Financial Derivatives Pricing

April 8, 2026

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1. If $f : [a, b] \to \mathbb{R}$ satisfies $|f(x) - f(y)| < K|x - y|$ for all $x, y \in [a, b]$, then f is of bounded variation.

2. Find the total variation of the following functions over the interval $[0, 2\pi]$

   (a) $\sin x$

   (b) $\cos x$

   (c) $\sin x + \cos x$

3. For the function

$$f(x) = \begin{cases} \frac{1}{1-x} & \text{if } x \neq 1 \\ 0 & \text{if } x = 1 \end{cases} \tag{1}$$

   find the total variation on interval $(0, 1)$ and $[0, 1]$.

4. The function f defined by

$$f(x) = \begin{cases} 0 & \text{if } x \text{ is irrational} \\ 1 & \text{if } x \text{ is rational} \end{cases} \tag{2}$$

   is not of bounded variation on any interval.

5. For the function

$$f(x) = \begin{cases} 1 & \text{if } x \in [0, 1) \\ 2 & \text{if } x \in [1, 2) \\ 3 & \text{if } x \in [2, 3] \end{cases} \tag{3}$$

   find the total variation on the interval $[0, 3]$.